

The Nature of Time: A Discussion on the Definition of Time, the Irreversibility of the Thermodynamic Arrow of Time, Subjective Time, Objective Time, Local Time, Proper Time Dilation and Cosmic Time

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Abstract:The nature of time is a profound and complex topic that involves multiple fields, including physics and philosophy. In this paper, time is defined as a discrete process related to the counting of state transitions associated with increasing entropy, where the Planck time is the smallest unit of time, and the direction of the arrow of time aligns with the direction of entropy increase. This understanding of time not only reflects the irreversibility described by the second law of thermodynamics but also explores the distinction between subjective and objective time. The concept of local time is also introduced, representing the time associated with a specific local space, and it is shown to be consistent with the concept of time in special relativity. Through discrete counting operations, the Lorentz covariance naturally emerges, explaining relativistic effects such as time dilation. Cosmic time, on the other hand, refers to the expression of time on a much larger scale, considering the overall evolution of the universe. These different levels of time concepts together form a multidimensional picture of time, offering a new perspective on the fundamental nature of time.

This is an introduction to the theory of time in the Draft titled “Time-Entropy Mapping; Mass-Gravity Duality; Metric-Frequency Mirroring” Preprints.
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The Nature of Time:

What is time? How do we understand time? How do we define time? These are frequent questions on Q&A platform, and they are also questions that philosophers, physicists, scholars from interdisciplinary fields, and ordinary people have repeatedly contemplated and pursued. It can be said that understanding time is a fundamental philosophical and physical question.

Throughout history and across cultures, philosophers and physicists have offered various perspectives on the nature of time:

Subjective theory of time: Some philosophers emphasize that time is merely the human consciousness perceiving changes in the external world. From Parmenides, Kant, to Husserl, these thinkers have highlighted the relationship between time and human consciousness.

Objective theory of time: Some philosophers and physicists emphasize that time is a count of motion or change processes. **Time-as-background theory:** This view considers time as an independent variable. Some modern physicists hold this view, as did Newton and others in classical physics. In modern quantum field theory, time is similarly treated as a background parameter, although some exploratory branches of quantum field theory consider time as a relational variable within a quantum system.

Time as the process of change in things: This understanding follows the objective theory of time. Modern thinkers such as Leibniz, Whitehead, and contemporary causal set theory all regard time as a collection of events and their relationships.

Spacetime unification: General relativity introduced the concept of spacetime as a unified whole, emphasizing the co-evolution of space and time. General relativity also introduced the idea that the metric field modulates local proper time.

Programming assigns time parameters to events, marking the sequence of events.

Irreversibility of time: Modern thermodynamic studies have revealed that change in real-world phenomena is at least macroscopically irreversible—this is known as the thermodynamic arrow of time or the law of entropy increase. Considering the event-set view of time, the law of entropy increase provides a fundamental driving force behind causal relationships between events.

Locality and non-locality of time and space: Both quantum field theory and general relativity traditionally treat time and space as local, limited by the speed of light, with event relationships following spatial and temporal adjacency. However, some contemporary theories, such as quantum entanglement theory, hypothesize the existence of non-local interactions, supported by a series of experiments.

Now, let's discuss the title of the article: What is the Nature of time?

The following are my views:

a. In fact, all theories of time share a common point: time is a measure or count of change—it is a scale or count of change, whether in a continuous or discrete view of change, whether subjective or objective. Subjective time is human perception of change, while objective time is the count of change in the universe as a whole or in its parts.

b. Only a theory of time that incorporates the thermodynamic arrow of time can be considered a good philosophical or physical understanding of time, because the law of entropy increase has been empirically verified, and no counterexamples have been found yet, whether at the microscopic or macroscopic level. Even at the QED level, even a single electron (there is no truly stationary electron) generates entropy through

radiation.

c. From the perspective of the rate of change, general relativity and related observational experiments have shown that gravitational fields indeed modulate the frequency of change in local regions—time slows down. That is to say, time is not an independent variable but a relational variable dependent on spatial change. When space is distorted or disturbed by energy, the rate of change in the corresponding local region also changes.

d. The evolution of things and the locality of space and time in their interactions form the foundation of contemporary physics and are also supported by empirical observations. Here, I want to discuss the issue of non-locality: if experiments like those violating Bell's inequality at low energy levels lead to non-local interactions in spacetime, then the laws of physics would not be what we currently observe. Moreover, the object of physical investigation would be an extremely complex system with a state space larger than the computational capacity of the universe itself. From this perspective, non-locality is not valid.

My view: My perspective is close to that of Einstein, Leibniz, Whitehead, and contemporary causal set theory. I believe that time is the count of cosmic change, the refreshment of cosmic states, the set of events, and the set of transformations. In fact, this understanding is easily accepted by professionals in the computer science field. At the same time, time is not an independent variable but is influenced by the change in space itself, as has been verified by observations and experiments, such as gravitational time dilation and gravitational redshift. Furthermore, I believe that space is quantized, and time is the set of spatial transformations. A transformation set of discrete sets is still a discrete set. To fully understand time, we must also distinguish between local time and cosmic global time. The time dilation in special relativity actually reflects different observers in different motion frames counting changes differently, rather than a concept of objective proper time. The gravitational time dilation in general relativity, however, describes how spatial distortion affects the local rate of change.

The following is the Sections of my preprint article on the nature of time. By constructing a multiplicative entropy-time-space transformation set with multiple mappings, I attempt to embed the thermodynamic arrow of time—its irreversible nature of energy distribution uniformization—into both general relativity and quantum field theory.

1 Space Configuration:

1.1 The universe is expanding.

1.2 The universe operates under the Law of Energy Conservation and the Law of Entropy Increase.

1.3 The universe consists of Space Elementary Quanta (SEQ), discrete Planck-scale units, forming

a topologically homeomorphic 3D structure (adjacency relations remain preserved while individual SEQ energy states may vary). Importantly, this polycrystalline structure necessarily contains primordial Periodic or Random Topological Dislocation to ensure physical isotropy, yet maintains strict 3D topological homeomorphism as these defects are cosmologically frozen and adjacency-preserving since the birth of the universe. The distortion of light around black holes demonstrates that gravitational and electromagnetic field quanta are coupled, suggesting they originate from the same quantum field in different excited states—leading to the SEQ hypothesis. The spacing and tension between adjacent SEQ can be modulated by gravitational or equivalent gravitational fields.

1.4 All field quanta and elementary particles represent different energy excitation states of SEQ, expressible as 3D dynamic structural matrices of SEQ .

1.5 SEQ possess a ground state energy (e.g., ground-state spin or vibration modes). If ground-state spin chirality is fixed, this may explain parity violation. The ground-state energy of SEQ could also account for the cosmological constant in General Relativity. This framework shows strong alignment with Loop Quantum Gravity theory.

1.6 Adjacent SEQ maintain a dynamic equilibrium spacing interconnected via spring-like bonds in their ground state. SEQ are fundamental, indivisible Planck-scale entities—their structure remains intact under any deformation or energy fluctuations.

The sub-Planckian regime governs the spatial elasticity of the network (including elastic potential storage and release), while quantized energy transfer occurs exclusively through interactions between SEQ. Within this framework:

- The spin degrees of freedom of SEQ and their elastic bonds remain decoupled, preserving independent dynamical regimes.
- Under perturbation, the system responds by modifying SEQ resonant frequencies while generating compressive/tensile forces.
- This elastic response is nonlinear and asymmetric.
- SEQ are stable, indivisible structures composed of sub-Planckian components. SEQ' spin emerges from collective space transformations at the sub-Planck level. This ensures the spin degrees of freedom do not interfere with elastic deformations in the SEQ network. This architecture naturally protects spin dynamics from elastic disturbances.
- At the sub-Planckian scale, the elastic properties of the underlying substrate impose an upper bound on the spacing modulation and tension between adjacent SEQ. This fundamental limit ensures that extreme deformations (e.g., near black hole singularities) cannot disrupt the topological integrity of the SEQ network.
- In this model, the harmonic oscillation intervals of SEQ are integer multiples of Planck time(t_p). Consequently, all dynamic processes—including elastic strain interactions, harmonic conduction, as well as scalar, spinor field transmissions and other energy conduction mode induced by rotational axis dynamics—are fundamentally constrained by the discrete Planck-time intervals. This property inherently ensures the model's consistency with the discrete-time hypothesis in quantum mechanics and quantum gravity theories.
- Any discrete model of spacetime must confront the challenge of restoring spatial isotropy so as to remain compatible with the Lorentz-covariant rules established by observation. Beyond the isotropy mechanism tied to the topologically dislocated configuration discussed in § 1.3, an alternative is to let the SEQ lattice spacing be sufficiently large for sub-Planckian elastic

constituents—whose characteristic scale is far below the Planck length—to fill the network uniformly. Provided that, within the precision accessible to cosmological observations, the statistical distribution of these constituents yields a dispersion relation that is effectively Lorentz-covariant, macroscopic isotropy emerges naturally and remains consistent with all current observational data.

1.7 If matter truly traversed space, it would require modification of spacetime's adjacency relations. Yet black holes—despite their extreme mass—preserve local spacetime topology (as evidenced by smooth light bending). This implies that apparent particle motion must instead represent propagating excitations of spacetime itself, consistent with GR. The speed of light (c) constitutes the maximum excitation propagation rate in space.

1.8 Matter-Spacetime Unity: Matter is a part of space

General Relativity tells us that matter affects the metric of space, and the metric of space in turn determines the motion of matter along geodesics. In other words, matter and space influence each other.

Algebraic module theory and category theory further reveal that two entities capable of mutual interaction must share part of the same algebraic structure. In categorical terms, relations are morphisms and interactions are structural.

Expressed in physical language, this means that matter and space must share a common underlying basis—hence, matter and space-time are one.

1.9 Quantum-Elastic Two-Layer Fiber Bundle Model | Common Base Space Platform for General Relativity and Quantum Field Theory

This quantum-elastic base space model, where the underlying structure consists of elastic space components at sub-Planckian scales, forming a topologically homeomorphic spatial network. Within this network, structurally stable space elementary quanta (SEQ) and nodal quantized spatial networks are constructed from sub-Planckian components. The SEQ are interconnected via sub-Planckian elastic components, forming a dual-layer fiber bundle base space structure, where the sub-Planckian components serve as the foundational substrate and the space elementary quanta form the secondary layer.

On this base space, due to the fact that the space elementary quanta themselves are composed of sub-Planckian components, the spin of these quanta is essentially a topologically protected spin excitation conduction wave on the sub-Planckian topological network. This allows for the decoupling of spin and elasticity without altering the fundamental spatial topology. Matter is fundamentally a compressed excitation state of space, and the motion of matter is essentially a topologically protected excitation wave on space. The motion at each level is essentially a wave on the substrate of the lower level, thus preserving the topological homeomorphism.

Based on this structure, it becomes straightforward to define two fundamental connections: the gauge group of quantum field theory (QFT) and the metric tensor of general relativity (GR). This enables the model to simultaneously accommodate the mathematical frameworks and phenomenological interpretations of both quantum field theory and general relativity.

Between these two connections, there also exists a categorical morphism: the geometry-frequency mirror, which will be discussed in Section 13.

In this quantum-elastic spatial network model, the energy carried by space elementary quanta can represent the quantum spin-resonance energy of space. Additionally, the elastic potential

energy of space can be decomposed and calculated across relevant regions of the space elementary quanta with the mirroring principle between geometric deformation and quantum resonance frequency introduced in section 13. Thus, the entire spatial energy distribution can be represented solely by the energy carried by the space elementary quanta.

Once the energy distribution across space is constructed using the energy carried by the space elementary quanta, each step in the dynamic energy distribution of this quantum-elastic space corresponds to each step of spatial transformation. The second law of thermodynamics manifests as a process of energy distribution homogenization. Based on this, multiplicative entropy can be used to define the entropy value of each spatial transformation, which can then be mapped onto the time set. This provides a clear definition of cosmic time and embeds the thermodynamic time arrow of the second law of thermodynamics into the models of general relativity and quantum field theory.

2 Time as a counting process of spacetime network transformations:

2.1 SEQ serve as the electromagnetic wave conducting medium. Matter with mass and its motion are waves in this medium. In this framework, nothing truly moves through space - light speed c is the maximum conduction speed c , preventing velocity stacking beyond c . All physical phenomena correspond to specific energy state configurations, establishing SEQ as the universal substrate.

2.2 The universe's composition: Energy conservation and quantization imply a finite number N of SEQ, each with M possible energy states, (where each energy state m_i is an integer multiple of Planck's constant h ,) allowing up to M^N transformations. These M energy states form an algebraic system incorporating translational, spinning and rotational operations connecting to standard model. Energy conservation and entropy increase constraints reduce possible transformations significantly below M^N .

2.3 Time definition:

2.3.1 Let J be the possible universe transformations ($J \ll M^N$).

2.3.2 The Planck time (t_p) interval separates adjacent transformations as the minimal time unit.

2.3.3 Time's arrow follows entropy increase.

2.3.4 Transformations map non-bijectively to entropy values (k distinct values partition J transformations into K classes). Parallel transformations share the same entropy values, but only one can occur. The entropy set maps to possible time values - each moment corresponds to one universe transformation. Non-uniform entropy increase means only a subset of possible time values actually occur.

2.3.5 Each space transformation (state transition of the SEQ network) can be assigned a unique entropy value calculated via the multiplicative energy distribution across this space transformation's matrix.

2.3.6 Finite transformations ensure discrete, limited time in this model.

(Note: this derivation is a speculative exploration within the discrete framework and does not imply reality.)

3 Definition and analysis formula of entropy In this definition of entropy, the entropy value of a closed system at a given moment (i.e., during a specific state transition) is calculated as the multiplicative product of the energy norms of all SEQ involved in that transition (that

moment's space transformation).

Entropy value of closed system $S = \prod m_i, i \in N$ ①

Energy of closed system=constant $= \sum m_i, i \in N$ ②

$S_{max} \leq m_i^n$, When all m_i are equal or differ only by Planck's constant h

(Where m_i refers to the energy carried by the i th SEQ during a single transformation of the closed system, where each energy state m_i is an integer multiple of Planck's constant h , $m_i=n_ih$, $n_i \in N$)

Each SEQ carries energy m_i = the resonant kinetic energy of this SEQ+its spin kinetic energy+the elastic potential energy of the local space surrounding the SEQ, calculated based on the gradient of resonance frequency shift relative to the Planck frequency of the SEQ, as will be elaborated in Section 9.1 and Section13.

Under this setup, the total energy of the entire space can indeed be represented solely by m_i , without the need to separately calculate the elastic potential energy within the sub-Planckian components.

3.1 Energy transfer rules and triggering conditions:
Energy exchange occurs between adjacent SEQ (i,j) if and only if the following thermodynamic gradient exists: $m_i > m_j+h$, Energy transfer occurs only in discrete quanta of Planck's constant h , $m_i \rightarrow m_i-h$; $m_j \rightarrow m_j+h$ (Planck's constant: h)

3.2 Numerical Example: System States and Entropy Evolution

Table.1: Simplified Entropy Increase Demonstration

System State	SEQ Energy Distribution $E_{total}=\sum m_i=12$	Entropy $S=\prod m_i$	Remarks
Initial non-equilibrium state	[3, 1, 5, 3]	45	-
Intermediate state	PathA[3, 1, 4, 4]; PathB[3, 2, 4, 3];	A48; B72;	-
Final state	PathA[3, 2, 3, 4]; PathB[3, 3, 3, 3];	A72; B81;	Due to adjacent energy transfer with minimal quanta h , this system cannot reach maximum entropy in case A

Note:The above analysis demonstrates that different entropy-increasing pathways exhibit distinct sequences of entropy variation.

3.3 Logarithmic Relation:
After logarithmic transformation, $\ln S$ aligns with the conventional Boltzmann entropy form, while the multiplicative formulation naturally suits discrete systems.

3.4 Proof of Spontaneous Entropy Increase
Spontaneity Theorem of entropy increase (Second Law of Thermodynamics):

For every possible energy transfer process, the total entropy change satisfies $\Delta S \geq 0$.

Proof Outline: Let the pre-transfer states be $m_i=a$, $m_j=b$ ($a > b+h$);

The post-transfer entropy ratio is:

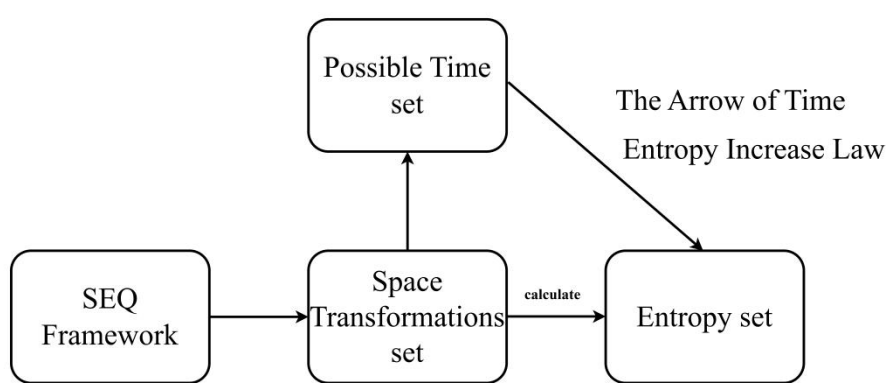
$$S_{t1}/S_{t2}=(a-h)(b+h)/ab=1+h(a-b-h)/ab>1 \text{ (Planck's constant } h)$$

3.5 The definition conserves energy, has an entropy ceiling, ensures spontaneous increase, and logarithmically aligns with classical entropy.

Indeed, this model establishes a novel analytic quantum thermodynamic framework where coarse-graining automatically enforces the indistinguishability of quantized homogeneous invariant network in physical properties while simultaneously characterizing the entropy-increasing process of energy homogenization. These properties collectively confirm the validity of this entropy definition. At this point, a clear multiple mapping can be established:

One space transformation→one entropy value→One possible moment

Space-Time-Entropy Mapping Diagram



Draw tool: draw.io [Computer software]. <https://github.com/jgraph/drawio>

Diagram.1: Space-Time-Entropy Mapping Diagram

3.6 The Spontaneous Entropy Increase is Causality.

The existence of isoentropic transformations implies that even with the constraint of the least action principle further limiting the degrees of freedom in future spatial transformation paths, the potential spacetime evolution trajectories are not necessarily unique. Within this model's framework, multiple solutions may satisfy the least action condition, indicating that while causality adheres to Markovian properties, a certain dynamic freedom persists.

The existence of isoentropic transformations and different entropy-increasing pathways imply that even with the constraint of the least action principle further limiting the degrees of freedom in future spatial transformation paths, the potential spacetime evolution trajectories are not necessarily unique. Within this model's framework, multiple solutions may satisfy the least action condition, indicating that while causality adheres to Markovian properties, with a certain dynamic freedom persists.

3.7 **Why Analytic Entropy** is Adopted: Within this model's framework, physical states at Planck-time scales exhibit deterministic characteristics. Although current experimental conditions cannot directly measure them, traditional statistical entropy can be reduced to analytic expressions

at this scale. This perspective not only reveals the microscopic essence of statistical quantities but also provides a new analytic foundation for quantum thermodynamics — unifying macroscopic statistical behaviors with deterministic dynamics at the Planck scale.

Multiplicative analytic entropy corresponds to each step of energy homogenization in a system. With every step toward a more uniform energy distribution, the multiplicative analytic entropy increases, offering a clearer and more detailed characterization of the process compared to traditional statistical entropy.

Multiplicative Entropy vs. Traditional Statistical Entropy

Comparison Dimension	Multiplicative Entropy	Traditional Statistical Entropy
Process Explicitness	Explicitly records energy homogenization steps via product sequences (e.g., $\prod_i m_i$), preserving microstate transition details	Describes only macro-state differences via logarithmic state-count ($\ln \Omega$), erasing intermediate dynamics
Physical Intuitiveness	Entropy increase directly reflects irreversible energy redistribution; time asymmetry emerges from dynamics	Relies on probabilistic assumptions (e.g., molecular disorder) and requires ad hoc low-entropy past boundary
Process Resolution	Tracks Planck-timescale (t_p) energy transfers;	Limited to ensemble averages, incapable of resolving quantum fluctuations or short-timescale entropy production

3.8 Analysis of the Maximum Entropy Principle

In the SEQ model, the maximum entropy principle is manifested through the driving tendency of entropy increase: energy not only flows from a higher-energy SEQ to an adjacent lower-energy SEQ, but it also follows the path with the largest energy difference.

For example, consider an SEQ i , which is adjacent to two other SEQ j and k . The energies carried by nodes i , j , and k are A , B , and C respectively, where $A > B > C$. In this case, there are two possible energy transfer paths from SEQ i according to the principle of entropy increase: path $i \rightarrow j$ or path $i \rightarrow k$. We will analyze the entropy change for each path separately.

Under the constraint of energy conservation (i.e., $A + B + C = \text{constant}$), the entropy of the local system composed of these three SEQ before energy transfer is: $S = A \times B \times C$

Path $i \rightarrow j$: transferring energy to node j (which has relatively higher energy):

$$S'=(A-1)(B+1)C=(A-1)(BC+C)$$

Path $i \rightarrow k$: transferring energy to node k (which has lower energy):

$$S''=(A-1)B(C+1)=(A-1)(BC+B)$$

Here, "1" represents one unit of Planck constant h . Since $B > C \Rightarrow BC+B > BC+C \Rightarrow S''>S'$, the entropy increases more along the $i \rightarrow k$ path—that is, the path with the larger energy difference leads to a greater increase in entropy.

This deduction can be easily generalized to cases where the number of adjacent nodes is greater than two, so a detailed proof is omitted here.

However, it should be emphasized that the path with the maximum energy difference is not necessarily unique. Therefore, for states that have not yet occurred, the future still retains

sufficient degrees of freedom — the evolution is not entirely deterministic.

3.9 In section 9.1.1, we will continue to discuss the decomposition of energy carried by SEQ $\{m_i = K_{\text{resonant}} (K_{\text{spin}}) + U_{\text{elastic}}$ assigned to this SEQ from its adjacent elastic bonds manifested as frequency suppression.}, the relational expressions, and the supplementary description of the initial low-entropy state of the universe. Chapter 13 will provide a detailed explanation of the metric-SEQ resonance frequency mirror relationship under this model. With this setting, there is no need to observe the sub-Planckian constituent-level elastic potential energy or the metric tensor; instead, the regional elastic potential energy can be directly represented by the SEQ resonance frequency. Chapter 14 will mention that SEQ spin and resonance frequency mutually excite each other and change synchronously; therefore, SEQ spin kinetic energy can also be represented by SEQ resonant kinetic energy.

5 Local time , the proper time and relative time in Relativity

5.1 **local time.** As previously established, global time is defined by the transformations of the universe. This work introduces local time as an operational concept that: (1) provides correspondence with special relativistic time notions; and (2) enables precise specification of time scales for localized physical processes. Crucially, any measurable time parameter in physical calculations necessarily corresponds to transformations within a specific local space. This operational concept is designated as **local time**.

The local space scope must be unambiguously specified: either as the SEQ network along the physical process path or the connected region within the observer's light cone. This distinction mirrors the complementarity between Feynman's path-integral formulation and relativistic theory, avoiding conceptual confusion in prior works.

Local time: it can specify the time set corresponding to the transformations within a specific spatial range. In fact, from the perspective of this framework, the existing equations with t parameter in textbooks are actually the local time by default.

One example: the clock slowness effect in Relativity manifests as different transformation times in different local spaces.

Every local space transformation constitutes a part of the universal evolution. Global time progression (ΔT_{global}) does not necessitate synchronous local transformations, the state matrix of a given local space may remain invariant despite cosmic-scale changes ($\Delta t_{\text{local}} \approx 0$) when this local space undergoes no state transition.

5.2 **The proper time in Relativity** is related to local transformations count of physics process entities. (Section 9.1, Section 9.5.6 present the physical mechanism underlying proper time dilation in GR.)

5.3 Understanding on **Lorentz-covariant rules** in **Special Relativity** theory: The time perception of physical processes across distinct reference frames fundamentally corresponds to the observation of transformation counts. The observation time discrepancy between frames derives from the accumulated difference in their frame SEQ transformation counts. An observer measures another frame's time evolution through the differential transformation count ΔN , while he can't perceive their own transformation count N_0 . The observed ΔN is fundamentally governed by the dynamic light-path difference between the observer's frame and the moving reference frame of the measured object. Under the principle of non-additivity of light speed (c-invariance), this

formulation naturally derives Lorentz-covariant rules through counting operations.

- Key Distinction from GR Effects

While the mathematical derivation process aligns with standard special relativity textbooks—replacing continuous spacetime metrics with discrete counting operations—the physical interpretation differs substantially in conceptualization:

- SR Effects as Perceptual Phenomena

Time dilation and length contraction emerge purely as observer-dependent measurement consequences in SR

Originate from information transmission constraints via discrete light-signal counting

- Contrast with GR Mechanisms

Gravitational time dilation involves actual deformation of the SEQ network's transformation frequency

Equivalence principle effects require space compression as well (see §9.5)

5.4 Physical Meaning of Planck Time

In this framework, the Planck time (t_p) corresponds to the **fundamental harmonic period** of the SEQ network in its equilibrium state—the duration for a complete energy transfer (or harmonic oscillation period) between adjacent SEQ. This period defines the **theoretical minimum transformation cycle** (i.e., maximum transformation frequency) of the universe as a whole.

While gravitational fields or equivalent interactions can locally modulate the harmonic frequency by deforming the SEQ network (e.g., compression/stretching, as in gravitational time dilation), the global maximum theoretical transformation frequency remains anchored by t_p in the equilibrium state. However, due to the omnipresence of gravitational effects, the empirically observable maximum transformation frequency of space may marginally less than this theoretical limit.

9.5.6 Gravitational and Kinematic Time Dilation

All factors that induce metric variation, including mass (gravity), velocity, and acceleration (equivalence principle), compress or stretch spacetime locally, thereby modulating the transformation frequency of related space. This frequency suppression constitutes the fundamental mechanism of time dilation.